Seminar WS25 Fredholm Operator and Index Theory: Compact Manifolds with Boundary

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1 Overview

On the previous talk, we got acquainted with the index theorem for Dirac operators of Atiyah and Singer in its local form. The setting was that of a closed spin manifold X and a Clifford bundle [Mel93, §3] $E \to X$, that is, a vector bundle which is a $Cl(X)$-module, i.e. the vector space E_x is a module over the algebra $Cl(X)_x$ for all $x \in X$. Assuming X is even dimensional and E is $\mathbb{Z}_2$-graded, we can consider the (generalized) Dirac operator associated to E:

$$D^{\pm} : \Gamma(X, E^{\pm}) \longrightarrow \Gamma(X, E^{\mp}), \quad \text{i.e.} \quad D = \begin{pmatrix} 0 & D^- \\ D^+ & 0 \end{pmatrix}$$

with respect to the grading $E = E^+ \oplus E^-$.

Theorem 1.1. [ABP73, §6][1]

$$\mathrm{ind}(D^+) = \int_X \hat{\mathrm{A}}(X)\mathrm{Ch}(E)$$

□

If we take the simplest case where $E = S$ is the spinor bundle, we already observe most of the interesting behavior, so we can for simplicity focus on this specific example (in that case, $\mathrm{Ch}(E)$ is trivial, so we just have the integral over the $\hat{\mathrm{A}}$-genus).

This theorem came as a unification of several previously known geometrical results. Two of the most relevant are[2]:

The Gauß-Bonnet-Chern theorem

Within an even dimensional oriented Riemannian manifold, we can extend the metric from vector fields to differential forms, and thus define the adjoint of the exterior differential d^*. Considering the vector bundle of differential forms of all degrees $E = \Lambda T^*X$, which can be splitted into even degree forms E^+ and odd degree forms E^-, it is clear that the operator $D = d + d^*$ maps $E^{\pm} \to E^{\mp}$.

[1] We take here the "topologists' normalization" of the $\hat{\mathrm{A}}$-genus, i.e.

$$\hat{\mathrm{A}}(X) = \det{}^{1/2}\left(\frac{\frac{R}{4\pi i}}{\sinh \frac{R}{4\pi i}}\right)$$

which simplifies the formula. We do the same with the L-polynomial. However, with the Pfaffian, we take the geometers definition, producing the factor $(2\pi)^{-n/2}$, so that the familiar form of the Gauß-Bonnet theorem $2\pi\chi(X) = \int_X K_X dvol_X$ is recovered.

[2] Both can be seen as index theorems for Dirac operators if the manifolds are spin, but these operators also exist in more generality and the theorems still hold.

The index of this operator is given by the Euler characteristic, and the local contribution of the index formula is in this case the Pfaffian [BGV04, 1.35, 4.7]:

$$\operatorname{ind}(d+d^*) = \sum_{i=1}^{n} (-1)^i H^i(X) =: \chi(X) = \frac{1}{(2\pi)^{n/2}} \int_X \operatorname{Pf}(-R)$$

In the case of surfaces ($n=2$), the Pfaffian corresponds to the Gaussian curvature K_X, which is half of the scalar curvature.

Hirzebruch's signature theorem

Let X be a closed oriented Riemannian manifold whose dimension n is a multiple of 4. Consider again the same operator and bundle as above, but with a new grading: the map

$$\tau : \Gamma(X, \Lambda^k TX) \longrightarrow \Gamma(X, \Lambda^{n-k} TX), \quad \alpha \mapsto i^{k(k-1)+\frac{n}{2}} \star \alpha$$

is an involution and so has eigenspaces corresponding to the eigenvalues ± 1 ($\star$ denotes the Hodge-star operator). We use them to define a grading for E, so that $E^{\pm}$ is the ± 1-eigenspace of τ. One can check that $D = d + d^*$ maps $E^{\pm} \to E^{\mp}$.
The signature $\operatorname{sign}(X)$ we are interested in is that of the quadratic form $H^{n/2}(X) \to \mathbb{R}$ given by taking cup product of differential forms representing the de Rham cohomology classes and evaluating against the fundamental class, i.e.

$$[\alpha] \mapsto \int_X (\alpha \smile \alpha)\, dvol_X, \quad \alpha \in H^{n/2}(X)$$

This is well-defined and depends on the choice of orientation: an orientation reversal swaps $(\tau, E^{\pm}, D, \operatorname{sign}(X)) \to (-\tau, E^{\mp}, D^*, -\operatorname{sign}(X))$.
The signature theorem realizes this signature as the index of the Fredholm operator D [ABP73, §5-6][BGV04, 4.8]:

$$\operatorname{ind}(d+d^*) = \operatorname{sign}(X) = \int_X L(X)$$

Our **goal** is to drop the assumption of closedness in the index theorem and generalize it to compact manifolds with boundary.
The Gauß-Bonnet theorem for surfaces with boundary is well-known:

$$2\pi\chi(X) = \int_X K_X dvol_X + \int_{\partial X} k_g dvol_{\partial X}$$

where k_g is the geodesic curvature of the boundary (which is a disjoint union of curves); this vanishes when the metric is of product type near the boundary[3]. That is, by allowing for boundaries we obtain a new summand only dependent on the boundary, and the rest is analogous to the closed case.
In the signature case, we can start by computing $\operatorname{sign}(X)$ and $\int_X L(X)$ in examples and rapidly noticing that they are not equal, even if the manifold is of product type near the boundary. In fact, the difference between them again only depends on the boundary, as the following remark shows.

[3]By this we mean that in a collar neighborhood of the boundary $\partial X \times [0, \delta)_u$ (which always exists) the metric takes the form $du^2 + g_{\partial X}$, where $g_{\partial X}$ is a metric on ∂X, with no u-dependence.

Remark 1.2. Consider two compact oriented manifolds X, Y with isometric boundaries $\partial X \cong \partial Y$. We can glue both manifolds together along their boundary by reversing the orientation of one of them, so that the boundaries fit, e.g. $X \cup_\partial (-Y)$. The orientation reversal has the effect $dvol_{-Y} = -dvol_Y$, $\operatorname{sign}(-Y) = -\operatorname{sign}(Y)$. The glueing produces a closed oriented manifold, which satisfies the signature theorem[4]:

$$\operatorname{sign}(X) - \operatorname{sign}(Y) = \operatorname{sign}(X \cup_\partial (-Y)) = \int_{X\cup_\partial(-Y)} L(X \cup_\partial (-Y)) = \int_X L(X) - \int_Y L(Y)$$

In particular, the difference $\alpha(\partial X) := \operatorname{sign}(X) - \int_X L(X) = \operatorname{sign}(Y) - \int_Y L(Y)$ is independent of the manifold which ∂X is the boundary of. It sometimes goes by the name of **boundary/signature defect**.

□

From the properties above we can deduce that $\alpha(\partial X) = -\alpha(\partial(-X))$ and in contrast to the right hand side of the index theorem, it is not a local quantity, since it does not behave multiplicative under coverings (vanishes on the sphere but does not for some lens spaces [APS75]). It is in fact a **global** spectral invariant.
Motivated by our previous treatment of the closed case, a possible strategy would be to set up a Fredholm problem for the signature operator above but in the boundary case. This requires us to choose boundary conditions.
If we do the same with the Gauß-Bonnet operator (recall: same as the signature operator but with a different bundle grading), we can set up a Fredholm problem by choosing local boundary conditions like Dirichlet or Neumann. This is not the case for the signature operator (there is an obstruction to Fredholmness with local boundary conditions that manifests as a non-vanishing K-theoretical class). The solution proposed by Atiyah-Patodi-Singer [APS75] is to choose a certain type of **global** boundary condition[5], which we illustrate with the following example:

Example 1.3. Let $\mathbb{D} \subset \mathbb{C}$ be the unit disk seen as a submanifold of the complex plane with coordinate $z = x + iy$. The boundary of $\overline{\mathbb{D}}$ is the unit circle $\mathbb{S}^1$. In the complex (Kähler) world, the Dirac operator coming from Hodge theory (i.e. the analogue of the above) is the Dolbeault operator $\bar\partial = \frac{1}{2}(\partial_x + i\partial_y) = \partial_{\bar z}$. Consider it acting on smooth sections of $\overline{\mathbb{D}}$. Due to the Cauchy-Riemann equations, its kernel corresponds to the space of holomorphic functions on the disk. That means, the restriction of the kernel is given by those smooth functions whose $\mathbb{S}^1$-Fourier series is of the form

$$u\big|_{\partial\overline{\mathbb{D}}}(z) = \sum_{n\geq 0} a_n z^n, \quad a_n = \frac{1}{2\pi}\int_0^{2\pi} u(x)e^{-inx}dx$$

that is, only non-negative powers are allowed. Note that the a_n are constructed from **global** information on the boundary.
As can be seen through integration by parts, the adjoint operator looks like $\bar\partial^* = -\frac{1}{2}(\partial_x - i\partial_y) = -\partial_z$. Its kernel contains the anti-holomorphic functions, where polynomials on $\bar z$ live. Their restriction of them corresponds to Fourier series with non-vanishing coefficients $n \leq 0$, since $\bar z^k = z^{-k}$ on the circle. The domain of this operator is adjusted to ensure the boundary integral in the integration by parts vanishes.
The free choice of the coefficients a_n tells us that the $\ker\bar\partial$ is infinite dimensional; the same is true for $\ker\bar\partial^*$. To tame them, we need a boundary condition. Dirichlet boundary conditions, together

[4]The first equality following from a Mayer-Vietoris argument and the third using the locality of the L-polynomial e.g. as pointwise supertrace of the corresponding heat kernel.

[5]With the boundary condition being necessarily global on the boundary, it is perhaps less surprising that the resulting spectral invariant also is global on the boundary.

with holomorphicity, force the function to be zero in the whole disk in the first place, due to the maximum modulus principle (the modulus of a holomorphic function on the disk has to be maximal on the boundary circle), giving $\ker \bar\partial = \{0\}$. But it also translates to no condition on the domain of $\bar\partial^*$, so its kernel stays infinite-dimensional and the operator is not Fredholm.
To remedy this and having this Fourier series in mind, a more appropriate boundary condition is to ask the Fourier coefficients a_n to vanish for $n > a$ for some choice of $a \in \mathbb{R}$. With this, $\ker \bar\partial = \lfloor a \rfloor + 1$ if $a \geq 0$ and zero otherwise, i.e. it can take any (non-negative) value but is finite dimensional. This is however a **global** boundary condition, since it is a requirement on the a_n's, which are defined globally on the boundary. The dual boundary condition on $\bar\partial^*$ is the vanishing of a_n for $n \leq a$ (cf. [APS75, (2.4)]), producing $\dim\ker \bar\partial^* = -\lfloor a \rfloor$ if $a < 0$ and zero otherwise. As a result, the problem is Fredholm with index:

$$\mathrm{ind}(\bar\partial) = \begin{cases} \lfloor a \rfloor + 1 & a \geq 0 \\ \lfloor a \rfloor & a < 0 \end{cases}$$

□

The setting in [APS75] is then the following: we consider a compact manifold with boundary X whose metric is of product type near the boundary[6] and a first order elliptic differential operator which on a collar neighborhood of the boundary $\partial X \times [0, \delta)_u$ takes the form:

$$D = \sigma(\partial_u + A)$$

Here, σ is the principal symbol, which is a bundle isomorphism due to ellipticity, and A is a self-adjoint elliptic first order differential operator acting on the closed manifold ∂X. Thus, there is a complete orthonormal basis of eigenvectors $A\phi_\lambda = \lambda\phi_\lambda$ which allow us to formulate a Fredholm problem[7]:

$$\begin{cases} Du = 0 \\ P_{\mathrm{span}\{\phi_\lambda : \lambda \geq 0\}} u = 0 \end{cases}$$

where P_H denotes projection onto H. [APS75, §2-3] show that this is in fact a Fredholm problem and compute the boundary defect to be the **η-invariant**:

$$\eta(\partial X) := \frac{1}{\sqrt{\pi}} \int_0^\infty \mathrm{Tr}\left(Ae^{-tA^2}\right) \frac{dt}{\sqrt{t}}$$

This quantity measures *spectral asymmetry* of the boundary operator A, in that it corresponds precisely with the value at $s = 0$ of the meromorphic extension of the function[8]

$$\eta_A(s) := \sum_{\lambda \in \mathrm{Spec}A \setminus \{0\}} \frac{\lambda}{|\lambda|^{s+1}} = \sum_{\lambda \in \mathrm{Spec}A \setminus \{0\}} \mathrm{sign}(\lambda)|\lambda|^{-s}$$

which is (a priori) holomorphic for $\mathrm{Re}(s)$ sufficiently large.

[6] In the non-product type case, the signature theorem contains a contribution similar to that of the geodesic curvature in Gauß-Bonnet.

[7] Note the analogy with the restriction on the Fourier series modes in the example.

[8] Suppose A is a finite rank operator, so that the trace above can be written as:

$$\mathrm{Tr}\left(Ae^{-tA^2}\right) = \sum_{\lambda \in \mathrm{Spec}A} \lambda e^{-t\lambda^2}$$

Then, carrying out the integral defining $\eta(\partial X)$ produces $\sum_\lambda \mathrm{sign}(\lambda)$ (use $\mathrm{sign}(\lambda) = \frac{\lambda}{|\lambda|}$ and the change of variables $s = \sqrt{t}|\lambda|$).

The idea for their proof relies on the fact that the collar neighborhood contribution to the trace of the heat kernel for a compact manifold with boundary has the same asymptotic expansion as $t \to 0$ as that of an infinite half-cylinder (with cross section ∂X). There is in fact a different perspective on this index theorem which results from considering the boundary as being unreachable [Mel93, §In.] or attaching the infinite half-cylinder just mentioned to the boundary [APS75, §3] and extending the operator naturally (i.e. to still have the same local expression above but now on $\partial X \times (-\infty, \delta)_u$). In that case, the equation

$$\sigma(\partial_u + A)f = 0$$

can be solved using the modes of A:

$$f = \sum_\lambda f_\lambda \phi_\lambda \implies (\partial_u + \lambda)f_\lambda = 0 \implies f_\lambda(u) = f_\lambda(0)e^{-\lambda u}$$

These allow us to understand the global boundary condition as an L^2-condition[9] on the cylinder:

$$\int_{-\infty}^{\delta} e^{-2\lambda u} du < \infty \iff \lambda < 0$$

We can moreover bring the cylindrical end to a finite spatial location ($x = 0$) by introducting the coordinate $x = e^u$. The metric and the operator then become:

$$g = \frac{dx}{x^2} + g_{\partial X}, \quad D = \sigma(x\partial_x + A)$$

near the boundary. This has a regular singular point at the locus of the cylindrical end $x = 0$ (meaning $(x - 0)$ multiplies the order 1 term of the operator).
To develop some intuition on why the η-invariant appears, we will consider the simplest case of a compact manifold with boundary in the next section. Before that, here is the complete index formula for a (generalized) Dirac operator D with induced boundary operator A:

Theorem 1.4. [APS75, §3]

$$\mathrm{ind}(D^+) + \frac{1}{2}\dim\ker A = \int_X \hat{\mathrm{A}}(X)\mathrm{Ch}(E) - \frac{1}{2}\eta(\partial X)$$

□

2 One-dimensional case

We follow the treatment in [Mel93, §1]. Consider the interval $X = [0, 1]$ and a first order differential operator $P = B(x)\partial_x + C(x)$ acting on sections of a bundle $E \to X$. The operator should be[10] adapted to the boundary structure and therefore should have regular singular points at the

[9] The explicit relation is given in [APS75, (3.11)] and is reflected in the index in [APS75, (3.14)] or [Mel93, p. 11].

[10] There are several intuitions why this is the correct class of operators to consider. One of them is that a one-dimensional manifold with boundary is modelled on $\mathbb{R}_+$ close to the boundary, whose Haar measure as a multiplicative group is $\frac{dx}{x}$ (if the manifold is closed, the model is $\mathbb{R}$ and the measure is dx, whereas the local Euclidean metric is dx^2). This measure corresponds to the Riemannian volume measure of the metric $\frac{dx^2}{x^2}$. The Laplace Beltrami operator of such a metric has a leading order term $x^2\partial_x^2$, and thus the corresponding Dirac operator has first order term $x\partial_x$. Because of P having this form on $[0, 1]$, it pulls back to the simple (and canonical) ∂_s on $\mathbb{R}$. On a similar vein, the role of the Fourier transform in studying pseudodifferential operators on closed manifolds is taken by the Mellin transform [Mel93, §5] for the boundary contributions, in the presence of boundary.

boundary loci $x = 0$ and $x = 1$. This amounts to $B(x) = x(1-x)\sigma(x)$. Ellipticity translates into $\sigma(x)$ being an invertible matrix for all x. We thus can write:

$$(Pf)(x) = \sigma(x)\,(x(1-x)(\partial_x f)(x) + A(x)f(x))$$

The boundary operator[11] is now the matrix $A(x)$, so the discussion above suggest that the boundary contribution to the index of a Fredholm problem in this context should involve its spectrum, i.e. its eigenvalues. Furthermore, the isomorphism $\sigma(x)$ plays no role in the index theory, so we can ignore it.

The simplest case is that of E being a line bundle, i.e. the matrices in P are all scalars:

$$P = x(1-x)\partial_x + a, \quad a \in \mathbb{C}$$

We want to find boundary conditions that give us a Fredholm problem (preferably with non-trivial index). As in the previous section, we can do this indirectly by choosing a domain for P that satisfies the appropriate L^2-regularity.

In this case, treating the boundary as unreachable would amount to pushing both ends to the limits of the real line. To achieve this, we can perform the change of coordinates:

$$x \in (0,1) \mapsto t = \frac{x}{1-x} \in (0,\infty) \mapsto s = \log t \in (-\infty,\infty) \implies P = \partial_s + a$$

It is now clear that the kernel of the operator contains functions of the form $f = ce^{-as}$, $c \in \mathbb{C}$. If we denote the operator $\partial_s + a$ by P_a, its adjoint is $P_a^* = -P_{-\bar{a}}$ (integration by parts).

A natural domain for P_a is the Sobolev space $H^1(\mathbb{R})$, which actually leads to the same domain for P_a^*. However, none of the functions in the kernel are square integrable, because either they grow exponentially at ∞ or at $-\infty$ (depending on the sign of $\mathrm{Re}(a)$), so the index is simply zero in this case.

Note that conjugating by exponentials produces operators of the same class[12]:

$$e^{-bs}P_a e^{bs} = P_{a+b} : e^{-bs}H^1(\mathbb{R}) \longrightarrow e^{-bs}L^2(\mathbb{R})$$

This means we can reduce the analysis to the case $a = 0$ by weighting the Sobolev spaces. After conjugating, the L^2-condition on the kernel becomes:

$$\int_{-\infty}^{\infty} c^2 e^{-2bs} e^{-2as} ds < \infty$$

which diverges around ∞ if $\mathrm{Re}(a+b) \leq 0$ and diverges around $-\infty$ if $\mathrm{Re}(a+b) \geq 0$. Again, we have no L^2-kernel, but the weighting has shifted the condition by $\mathrm{Re}(b)$. This inspires a solution: introduce different weights at $s = -\infty$ and $s = \infty$ (equivalently at $x = 0$ and $x = 1$) so that we obtain non-trivial kernel functions. From $e^{bs} = x^b(1-x)^{-b}$, we are motivated to study [Mel93, Def. 1.2]:

$$P_a : x^\alpha(1-x)^\beta H_b^1([0,1]) \longrightarrow x^\alpha(1-x)^\beta L_b^2([0,1]), \quad \alpha,\beta \in \mathbb{R}$$

[11] Since the manifold is odd dimensional here, there is no interior contribution to the index theorem. The boundary is 0-dimensional and therefore the bundle over it, where the boundary operator acts, is made up of just a vector space over $x = 0$ and a vector space over $x = 1$, so the operator is a matrix.

[12] Starting with P_a for which we fixed the domain $H^1(\mathbb{R})$

$$P_a : H^1(\mathbb{R}) \longrightarrow L^2(\mathbb{R}), \quad Pf = \partial_s f + af$$

then its conjugate by e^{bs} should inherit a weighted domain and range:

$$P_{a+b}f = e^{-bs}P_a(e^{bs}f) \implies P_{a+b} : e^{-bs}H^1(\mathbb{R}) \longrightarrow e^{-bs}L^2(\mathbb{R})$$

(so that $e^{bs}f \in \mathrm{dom}(P_a)$ and because $P_a(e^{bs}f) \in \mathrm{ran}(P_a)$).

where the subscript b means local integrability (like L^2_{loc} or H^1_{loc}) together with global integrability with respect to the measure $\frac{dx}{x(1-x)}$[13]. This **global** condition explicitly reads:

$$\int_0^1 |x^{-\alpha}(1-x)^{-\beta}f|^2\frac{dx}{x(1-x)} < \infty \iff \int_{-\infty}^{\infty} |(1+e^s)^{\alpha+\beta}e^{-\alpha s}f|^2 ds < \infty$$

When $s<0$, $1<1+e^s<2$, so the condition in this range is:

$$\int_{-\infty}^0 e^{-2\alpha s}|f|^2 ds < \infty$$

For $s>0$ we have $e^s < 1+e^s < 2e^s$ so we are essentially imposing:

$$\int_0^{\infty} e^{2\beta s}|f|^2 ds < \infty$$

showing how this in fact weights the integrability at each boundary point differently.
By conjugating we can reduce the problem to P_0 as mentioned, which shifts the weights of the Hilbert spaces ($(\alpha,\beta)\mapsto(\alpha+\mathrm{Re}(a),\beta-\mathrm{Re}(a))$, but we rename the new parameters by α and β to work without lose of generality). Using that functions in the kernel of P_0 are constant, we deduce from the integrability conditions on the domain (and because we can construct an explicit inverse, namely integration):

$$f\in\ker P_0 \subset \mathrm{dom}(P_0) \iff \alpha<0 \quad \text{and} \quad -\beta>0$$

In the case of the adjoint operator we obtain the same result up to reversing both signs. the result is:

$$\mathrm{ind}(P_a) = \begin{cases} 1 & \alpha < -\mathrm{Re}(a),\ \beta<\mathrm{Re}(a) \\ -1 & \alpha > -\mathrm{Re}(a),\ \beta>\mathrm{Re}(a) \\ 0 & \text{else} \end{cases}$$

i.e. α is worth $\frac{1}{2}$ if it is strictly smaller than $-\mathrm{Re}(a)$ and $-\frac{1}{2}$ if strictly greater, and similarly for β and $\mathrm{Re}(a)$. If equality happens in any case, the range is not closed and we lose Fredholmness.
The first step to generalize this result is to consider $a=a(x)$ to be non-constant in X. Easiest is if $a(x)=(1-x)a_0+xa_1$, for which

$$\frac{x^{a_0}}{(1-x)^{a_1}}P\frac{x^{-a_0}}{(1-x)^{-a_1}} = x(1-x)\partial_x$$

to which the result above can be applied. Notice that the conjugation now maps $(\alpha,\beta)\mapsto(\alpha+\mathrm{Re}(a_0),\beta-\mathrm{Re}(a_1))$, so

$$\mathrm{ind}(P) = \begin{cases} 1 & \alpha < -\mathrm{Re}(a_0),\ \beta<\mathrm{Re}(a_1) \\ -1 & \alpha > -\mathrm{Re}(a_0),\ \beta>\mathrm{Re}(a_1) \\ 0 & \text{else} \end{cases}$$

If E has rank greater than 1 and the boundary operator is of the form:

$$A(x) = (1-x)A_0 + xA_1$$

[13] This is the pullback by the change of coordinates of the Haar measure ds on $\mathbb{R}$ or $\frac{dt}{t}$ on $\mathbb{R}_+$.

where A_0, A_1 are diagonal matrices, constant on x, then we can treat coordinates independently. The operator acts on the j-th coordinate as $x(1-x)\partial_x + (1-x)A_0^{jj} + xA_1^{jj}$ (here A_i^{jk} is the entry of the matrix A_i in the row j and column k), thus it contributes to the index by:

$$\begin{cases} +1 & \alpha < -\mathrm{Re}(A_0^{jj}),\ \beta < \mathrm{Re}(A_1^{jj}) \\ -1 & \alpha > -\mathrm{Re}(A_0^{jj}),\ \beta > \mathrm{Re}(A_1^{jj}) \\ 0 & \text{else} \end{cases}$$

If we sum up the contributions from each coordinate, taking into account that the diagonal entries are precisely the eigenvalues, we get:

$$\mathrm{ind}(P) = \frac{1}{2}\left(\sum_{\substack{\lambda\in\mathrm{Spec}(-A_0)\\ \lambda>\alpha}} \dim E_\lambda - \sum_{\substack{\lambda\in\mathrm{Spec}(-A_0)\\ \lambda<\alpha}} \dim E_\lambda + \sum_{\substack{\lambda\in\mathrm{Spec}(A_1)\\ \lambda>\beta}} \dim E_\lambda - \sum_{\substack{\lambda\in\mathrm{Spec}(A_1)\\ \lambda<\beta}} \dim E_\lambda \right)$$

with E_λ the eigenspace corresponding to the eigenvalue λ of the matrix over which we are taking the sum. This holds again only when $\alpha \notin \mathrm{Spec}(-A_0)$ and $\beta \notin \mathrm{Spec}(A_1)$.
It turns out (solving the ODE) this is the behavior for a general P of the form we considered at the start, up to changing A_0 and A_1 by $A(0)$ and $A(1)$. In particular, if $\alpha = \beta = 0$ the problem is Fredholm whenever neither $A(0)$ and $A(1)$ have eigenvalue 0[14], i.e. whenever they are invertible, and the index is:

$$\mathrm{ind}(P) = \frac{1}{2}\left(\sum_{\lambda\in\mathrm{Spec}(-A(0))} \mathrm{sign}(\lambda) + \sum_{\lambda\in\mathrm{Spec}(A(1))} \mathrm{sign}(\lambda) \right) = \frac{1}{2}\left(\eta_{A(1)} - \eta_{A(0)}\right) = \mathrm{sf}(x \mapsto A(x))$$

so this is precisely the spectral flow of the family $A(x)$[DSBW23, (1.1)], also connecting this notion with that of the η-invariant![15]

References

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[14] If this approach is taken to study the problem of compact manifolds with boundary by attaching an infinite cylinder and looking for the appropriate regularity domains of the operators, the condition is also $0 \notin \mathrm{Spec}(A)$ [Mel93, (In.29)].

[15] This connection goes further, as described e.g. in [APS76, §7].